# Seloni Sinha

selonisinha2003@gmail.com linkedin.com/in/selonisinha selonisinha03

## Education

**Indian Institute of Technology (IIT) Dhanbad** Dec 2021 - May 2025
*Bachelor of Technology in Mechanical Engineering (GPA: 8.26/10)* *Dhanbad, Jharkhand*

- **Relevant Coursework:** Object-oriented programming, Data Structures and Algorithms, Building Software systems

**Stella Maris Inter College** April 2010 - April 2021
*10th Percentage:-99% (**ICSE ALL INDIA RANK 4**), 12th Percentage:-96.25%* *Lucknow, Uttar Pradesh*

## Experience

**Jaguar Land Rover** July 2025 – Present
*Graduate Software Engineer (Full Time)* *Bangalore, Karnataka*

**Ontology for Diagnostic Data**

- Collaborating with UK colleagues to develop a comprehensive **ontology design** that models key vehicle diagnostic data.
- Facilitating the integration of a **Virtual Knowledge Graph** with a JLR-specific **LLM-powered assistant**.
- Developing a unified ontology to standardise vehicle diagnostic data and concepts, enabling improved **data analysis** across platforms.

**Runbook Automation System**

- Developing an AI-powered runbook automation system to resolve incidents by executing predefined runbooks for triggered control points and error codes.

**Personalised In-Car Voice Assistant**

- Developing a personalised in-car voice assistant that leverages **user preferences** to deliver context-aware interactions.
- Working on a **category-based memory extraction** mechanism to identify, classify, and retain relevant user preferences.
- Leveraging extracted preferences to enable adaptive voice interactions and enhance the **in-car user experience**.

**Jaguar Land Rover** May 2024 – July 2024
*Data Science Internship (PPO received)* *Bangalore, Karnataka*

- Preprocessed a large-scale, multi-source image dataset captured from vehicle-mounted cameras, ensuring consistency in resolution and labelling for a **semantic segmentation** project.
- Implemented multiple **deep learning models** to enhance driver-assistance systems, focusing on real-time perception and scene understanding.
- Contributed to an R&D project focused on **optimising fuel efficiency** in vehicles during traffic conditions.

## Projects

**SummarEase** | *Python, AWS, GitHub Actions, Docker, FastAPI* *GitHub*

- Developed an end-to-end NLP pipeline for text summarization, leveraging transformers and deep learning models
- Implemented MLOps best practices with GitHub Actions for CI/CD, ensuring automated testing, model retraining, and deployment.
- Deployed the model using Docker, FastAPI and AWS, enabling scalable and efficient API integration
- Monitored model performance with MLflow and logging mechanisms, incorporating feedback loops for continuous improvement

**MedIntel** | *Python, TensorFlow, MLflow, Docker, DVC* *GitHub*

- Developed a deep learning-based kidney disease classification model using CNNs and traditional ML algorithms for accurate diagnosis
- Implemented MLOps best practices with MLflow for experiment tracking and DVC for dataset versioning, ensuring reproducibility
- Deployed the model using FastAPI, Docker, and AWS to enable real-time inference

- Automated CI/CD workflows with GitHub Actions, optimizing model retraining and deployment for continuous improvement

**RAGenius** | *Python, Llama2, FAISS, AWS Bedrock, LangChain, Boto3* *GitHub*

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system using Llama 2, AWS Bedrock, and LangChain for intelligent and context-aware responses
- Integrated LLM with FAISS to enable efficient document retrieval and intelligent response generation
- Built a serverless and scalable architecture using AWS Lambda, API Gateway, DynamoDB, and S3, leveraging Boto3 for AWS service interactions
- Automated MLOps workflows with GitHub Actions and AWS SageMaker, enabling continuous model improvement and seamless deployment

## Technical Skills

**Languages**: Python, C, C++, Javascript, SQL
**ML/Deep Learning**: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas
**LLMs and Agentic AI**: LangChain, LangGraph, Model Context Protocol(MCP), RAG, FAISS, AWS Bedrock
**Knowledge Graphs and Semantic Web**: OWL, SPARQL, RDFLib, Owlready, Ontop (OBDA/Virtual Knowledge Graphs), RMLMapper
**MLOps and Cloud**: Docker, FastAPI, MLflow, DVC, GitHub Actions, AWS (SageMaker, Lambda, S3, DynamoDB, API Gateway)
**Database and Tools**: MongoDB, Redis, Git/GitLab, Kiro

## Achievements

- **All India Rank 4** in ICSE Class 10 Board Examination among 2 million candidates
- **Top 1%** All India in JEE national exam
- **1st Rank** in National EV Thon competition in east India's largest tech fest
- **Competitive programming** : Global Rank 593 in CodeChef Starters; 400+ problems solved across CodeChef, Codeforces and LeetCode; 5 star on HackerRank

## Social Engagements

**Organizer at HackFest'23**: 36-hour long official hackathon organised by IIT (ISM) Dhanbad
**Technical Team Associate** : Steering and Suspension division at Electric Mobility Club, the official Formula Racing club of IIT (ISM) Dhanbad
**Operations and Sponsorship team member** : At Electric Mobility Club
**Active member at CODEISM**: The coding club of IIT Dhanbad
**School Representative** : Inter School Literary Competition